

The Multimedia Player's Assistant v1.0

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Overview

- Why this project was done
- Designing Phase
 - Testing, Surveying
- Construction Phase
 - Creating User Interface
- Implementation Phase
 - Features Involved
- Product Demonstration
- The Future
- Career Connection

Why this Project was Done

- May 2005
 - Wanted to design a way to organize certain tracks into a simple program.
 - MSDN Online Tutorial (How to Play Music using Visual Basic .NET)
 - Designed 4 user interfaces related to Specific Genre of Music and Video.

The Beginning



The Next Steps

- Test to see if a Data Set similar to a MS Access Table can be used as a type of library to store a collection of audio tracks.
- Test Windows Media Player ActiveX Component Capabilities.
- Create a Multimedia Player that is easy to navigate and has some extra features.

Testing Phases



The Survey

- Online Survey given to 10 students (via Blackboard) asking for their opinion on several possible features to include in this Multimedia Player.

The Survey (Questions)

- Compatibility for numerous multimedia files
- Database Playlist for organization
- Dual Media Players & Controls
- Crossfader
- Image Display with playing track
- Ordered, Random and Continuous Playback
- Customizable Interface (Background, etc.)
- Additional Suggestions

The Survey (Responses)

- Compatibility for numerous multimedia files
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User-Interface Construction

- **Major Problem**
Current Windows Media Player ActiveX Component would not recognize C-D Audio Tracks (.cda files)
- **Solution**
A newer version of the ActiveX Component (based on version 10) was able to read the C-D Audio Data.

User-Interface Construction



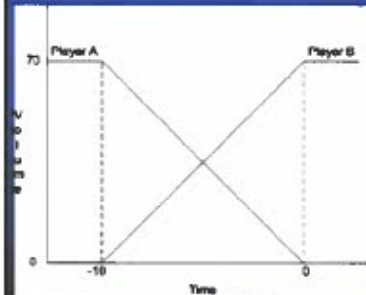
Data Table and Form



Implementation of Features

- Crossfader
- Playlist
- Automation for Crossfader and Playlist

The Crossfader



A simple timer-based function that reduces the volume for one track and increases the volume for the next track.

Operational either manually or automatically (When track ends)

The Crossfader

- User specifies how long the transition will take (1, 5, or 10 seconds)
- Number selected determines the speed of transitioning

$$\text{Speed} = \text{Number} \times \text{Time} / \text{Player Volume}$$
- Speed represents intervals for a timer function to continually repeat itself in transitioning (reducing volume by 1 unit).
- Crossfader will continue this transition until the previously played track volume reaches 0.

The Playlist

- User can add tracks from library or from files.
- Three Options for Playback
 - Ordered (Played in sequence [1st to Last] as user added them)
 - Random
 - Continuous (Same as Ordered but playlist starts over when the last track on the playlist ends)

Media Player Automation

- Composed of timer-based functions that oversee the status of the players.
- Some functions include:
 - Progress of track
 - Checking for when to transition
 - Checking for when to prepare the next track into another player

Program Demonstration

The Future of The Multimedia Player's Assistant

- Enhancements and Upgrades (v2.0?)
 - Redesign for Windows CE (Pocket PC)
 - More Playlist Capabilities
 - Streaming Audio (Internet)
 - File Conversions
 - Implementation in another language

A Career in VB.NET

- Employed at The Walt Disney World Resort
 - Information Management (Animal Programs)
 - Visual Basic .NET
 - Multimedia Design and Implementation
 - Data Collection for Animal Research & Technology Division (Bionformatics)

Resources Used

- MSDN Online Library
 - Windows Media Player Future Component Implementation
- Visual Basic .NET in easy steps (Programming for Windows and the Web) by Tim Anderson
 - Data Table & Form Implementation
- An Introduction to Programming Using Visual Basic .NET (5th Edition) by David F. Schneider
 - Other Functions and coding